precise: the so-called *AdS/CFT correspondence*, proposed by Juan Maldacena in 1997. The "AdS" in the label stands for "anti–de Sitter space," a hypothetical spacetime with no matter sources other than a negative vacuum energy (as opposed to the positive vacuum energy of our real world). "CFT" stands for *conformal field theory*, a particular kind of quantum field theory that can be defined on an infinitely faraway boundary of AdS. According to Maldacena, these two theories are secretly equivalent to each other. That's extremely provocative, for a couple of reasons. First, the AdS theory includes gravity, while the CFT is an ordinary field theory that has no gravity at all. Second, the boundary of a spacetime has one fewer dimensions than the spacetime itself. If we consider four-dimensional AdS, for example, that is equivalent to a three-dimensional conformal field theory. You couldn't ask for a more explicit example of holography in action.



Going into the details of AdS/CFT would require another book entirely. But it is worth mentioning that it is in this context that most modern research on the connection between spacetime geometry and quantum entanglement is being carried out. As noted by Shinsei Ryu, Tadashi Takayanagi, Mark Van Raamsdonk, Brian Swingle, and others in the early 2000s, there is a direct connection between entanglement in the boundary CFT and the resulting geometry in the AdS interior. Because AdS/CFT is relatively well defined as models of quantum gravity go, understanding this connection has been the target of a very intense effort over the past several years.

Alas, it's not the real world. All of the fun of AdS/CFT comes from relating things in the interior, where gravity happens, to things on the boundary, where gravity is absent. But the existence of the boundary is very special to anti–de Sitter space, which relies on a negative vacuum energy. Our universe appears to have a positive vacuum energy, not a negative one.

There's an old joke about the drunk who is looking under a lamppost for his lost keys. When someone asks if he's sure he lost them there, he replies,